

I. Benzene Structure and Resonance

Normal State: In its natural state, benzene possesses delocalized π electrons that form an electronic cloud above and below the ring, rather than being fixed in static double bonds.

Resonance Requirements: For resonance to occur, a specific sequence must exist: double-single-double bonds, a double bond separated by a single bond from a lone pair, a negative charge, or a positive charge.

Electron Movement: During resonance, the movement of electrons from one atom to another leaves a positive charge on the atom that lost them. Carbon atoms cannot remain negatively charged for long due to their lower electronegativity compared to atoms like nitrogen; therefore, they continue the resonance cycle to redistribute the charge.

II. Substituent Effects: Directing and Activating Groups

Electron-Donating Groups (Activators): Groups like -NH_2 have lone pairs that participate in resonance with the ring and increase electron density at ortho and para positions.

Electron-Withdrawing Groups (Deactivators): Groups like -NO_2 pull electrons away, creating positive charges at ortho and para positions, so electrophiles attack meta.

The Halogen Exception: Halogens are deactivators but direct to ortho/para due to resonance donation.

III. Aromaticity Requirements

Aromatic compounds must be cyclic, planar, and fully conjugated (all atoms sp^2).

Huckel's Rule: $(4n+2)$ π electrons.

Non-aromatic: Fail one of the main conditions.

Anti-aromatic: Meet conditions but have $4n$ electrons (unstable).

IV. Electrophilic Aromatic Substitution (SEAr) Reactions

Nitration: $\text{H}_2\text{SO}_4 + \text{HNO}_3 \rightarrow \text{NO}_2^+$.

Sulfonation: Adds $-\text{SO}_3\text{H}$.

Halogenation: Halogen + Lewis acid.

Friedel-Crafts Alkylation: Alkyl halide + Lewis acid.

Friedel-Crafts Acylation: Acid chloride or anhydride.

V. Critical Limitations and Exceptions in Friedel-Crafts

Deactivated Rings: No reaction with strong electron-withdrawing groups.

Strong Activators: $-\text{NH}_2$ / $-\text{OH}$ react with AlCl_3 and deactivate.

Rearrangement: Avoid by acylation then Clemmensen reduction.

VI. Nucleophilic Aromatic Substitution (SNAr)

Aryl halides do not follow SN_1 or SN_2 .

Addition-Elimination: Stabilized by electron-withdrawing groups.

Elimination-Addition: Via benzyne using strong base like NaNH_2 .

VII. Key Nomenclature and Intermediates

Phenyl vs. Benzyl: Benzyl has CH_2 attached.

Sigma Complex: Positive intermediate.

Allyl vs. Vinyl: Allyl (3C), Vinyl (2C).
